

Aviral Pathak

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EDUCATION

Madhav Institute of Technology and Science, Gwalior, *B.Tech in Mathematics and Computing* — 08/2025 – Present

Vishwa Bharati Public School, Noida — 04/2011 – 05/2024
CBSE Senior School Certificate Examination (Class XII)' 2024
Score: 88.2%
CBSE Secondary School Examination (Class X)' 2022
Score: 97%

PROFESSIONAL EXPERIENCE

Founder's Office (Technical Lead), *Vritam Technologies Pvt. Ltd.* — 07/2026 – Present, Gwalior, India
- Spearhead critical technical initiatives across **AI, robotics, and automation**, reporting directly to the Founder to align **product development** with strategic business goals.
- Architect and develop high-performance, **cross-platform mobile applications using *Flutter* and *Dart*** to provide intuitive control interfaces for **industrial automation systems**.
- Design and deploy custom **Machine Learning models** for ***predictive analytics*** and ***data processing***, enhancing the efficiency and intelligence of **core robotic solutions**.
- Streamline technical workflows and collaborate on system architecture, ensuring **robust integration between hardware components** and **AI-driven software interfaces**.

Campus Ambassador, *HackerRank* — 01/2026 – Present
- Setup and Created ***HRCC (HackerRank Campus Crew)*** in my college.

Student Builder Campus Leader, *AWS* — 08/2026 – 09/2026
- Spearheaded AWS initiatives at Madhav Institute of Technology and Science, fostering a highly engaged student **cloud developer community**.
- Organized impactful **technical workshops**, bridging academic curricula with industry-standard cloud deployment practices.
- Mentored peers in **core AWS services**, **guiding practical labs for deploying scalable full-stack web and machine learning applications.**

Co-Founder & Technical Lead (CTO), *Sunitiq-AI LLP* — 10/2025 – 09/2026
- Developed **EpiAlert-India's First Disease Outbreak Prediction System**

Research and Development Team Member, *AI Club, MITS Gwalior* — 09/2025 – 09/2026

Technical Committee Member, *GFG Campus Body MITS GWL-DU* — 09/2025 – 03/2026
- Organised two **DSA contests (Coderush and Coderush2.0)** with **GeeksForGeeks Campus Body, MITS-DU**

PROJECTS

SignStream, *Continuous Sign Language Recognition (CSLR) App*
- **Cross-Platform Architecture:** Engineered ***SignStream-CSLR*** using ***Flutter*** and ***React.js*** to ***translate continuous sign language into real-time text and speech***. Integrated ***MediaPipe*** and ***OpenCV*** for edge-based extraction of 3D skeletal coordinates from live video, eliminating high-bandwidth transmission latency.
- **Deep Learning & Sequence Modeling:** Developed a continuous translation pipeline in ***PyTorch*** to decode unsegmented spatio-temporal gestures. Trained bidirectional ***LSTMs*** utilizing ***Connectionist Temporal Classification (CTC) loss***, empowering the model to ***accurately interpret*** fluid, continuous hand kinematics over time.
- **Backend Database Management:** Architected an event-driven streaming backend integrating ***Node.js***, ***FastAPI***, and ***Socket.io*** . Leveraged ***PostgreSQL*** for robust ***relational data handling*** and vocabulary mapping, achieving sub-second inference latency to create an accessible, seamless communication tool.

CollabGrid, *Technical Matchmaker*
- **Cross-Platform Architecture:** Engineered ***"CollabGrid"***, a ***technical matchmaker***, utilizing ***Flutter*** and ***Dart*** to revolutionize how students ***form technical teams for hackathons and campus projects***. Implemented ***Riverpod*** for ***efficient state management*** and ***seamless interface rendering***.
- **AI-Driven Matchmaking:** Developed an intelligent, data-driven automated algorithm to replace random grouping by analyzing technical profiles ***highlighting specific expertise*** like ***Web Dev, ML, and Design***. Leveraged ***Scikit-learn*** and ***TensorFlow*** with ***NLP embeddings*** to ***accurately suggest compatible partners***, ensuring every team has a ***balanced mix of capabilities***.
- **Backend Database Management:** Architected a highly scalable backend integrating ***Node.js*** and ***PostgreSQL*** for robust ***relational data handling***. By bridging the gap between isolated talent and collaborative success, ***CollabGrid*** rapidly fosters innovation and creates a ***strongly connected campus ecosystem***.

Smart Attend, *Automated Attendance System for Rural Schools.*
- **Dual-Mode Tracking & AI Integration:** Architected a robust ***dual-mode attendance system*** utilizing ***OpenCV*** and ***TensorFlow*** to drive ***high-accuracy facial recognition***, supplemented by a ***hardware-integrated QR and RFID tag*** scanning module to ensure seamless operational redundancy.
- **Full-Stack Architecture & UI/UX:** Engineered a scalable ***web application*** utilizing ***React.js*** for the client-side and a ***Node.js*** runtime environment coupled with ***PostgreSQL*** for secure backend data management. Designed a ***clean, icon-based, and multilingual user interface*** optimized to guarantee ease of use for educators and ***students with minimal technical expertise***.

- **Scalable Rural Deployment:** Built specifically for r***esource-constrained rural educational environments***, prioritizing a highly flexible and scalable infrastructure. This innovative approach ensures a ***low-cost*** entry barrier while maintaining ***enterprise-grade system*** reliability and deployment efficiency

EpiAlert, *India's first Disease Outbreak Prediction System*
- By integrating heterogeneous data sources, EpiAlert aims to **detect rising outbreak risks earlier than traditional surveillance systems**, enabling proactive public health response.
- Made **EpiAlert Health Heroes** gamified dashboard under this project for surveying and filling forms

EXTRACURRICULARS AND ACHIEVEMENTS

- Student Representative in **Anti-Ragging Committee**
- **Endurance Race winner** at **Hindustan Formula Karting Championship (HFKC) - Season 6** held in Bhopal.
- Selected for **Smart India Hackathon (SIH)'26** in second round.
- **AICTE IDE Bootcamp** Winner, Represented **Sunitiq** in that bootcamp and came in **Top-5** teams
- Finalist at **HackIgnite 1.0**
- Winner of **AI-Unboxed**, a contest organised by **Soft Computing Research Society**
- Participated in **7 Hackathons**

SKILLS

Languages: C++, Dart, Python

ML and DS: Libraries: NumPy, Panda, Matplotlib, Seaborn | **Frameworks**: Scikit-Learn, PyTorch | **Concepts**: Supervised & Unsupervised Learning, Statistical Modelling, Data Preprocessing

Frameworks: Flutter, React.js, Next.js, Flexbox, Tailwind CSS, Bootstrap, MUI

Architectural: REST API, Widget Tree, Element Tree, Stateless vs. Stateful Widgets, Component-Based Architecture

Databases: PostgreSQL, SQL

State Management: Provider, Riverpod

Developer Tools: Git/GitHub, Android Studio, Figma(Basic), PSPP, Jupyter Notebook, Google Colab

HONORS AND AWARDS

SOF IMO 2023-Gold Medal Of Distinction, *Issued by Science Olympiad Foundation* 12/2023
Zonal Rank 1 in International Mathematics Olympiad 2023(SOF) Stage-1, got awarded with Medal of Distinction.

राष्ट्रीय हिंदी ओलिंपियाड(2019)-रजत पदक विजेता, *Issued by Hindi Olympiad Foundation* 11/2019
राष्ट्रीय हिंदी ओलिंपियाड (2019) प्रथम चरण में रजत पदक से सम्मानित

TEST SCORES

AIR 1165, *Got an All India Rank of 1165 in common merit list of B. Tech, as conducted by the Directorate of Technical Education, Madhya Pradesh.* 06/2025

AIR 67 (99.8 %ile), *Joint Entrance Examination(JEE) Main-Paper 2B (B.Planning)* 05/2025
(Uttar Pradesh State Topper)

AIR 666 (99.0961 %ile), *Joint Entrance Examination(JEE) Main-Paper 2A (B.Arch)* 05/2025

AIR 19, *GBUET-24* 06/2024